2025 IEEE International Conference on Data Science, Agents, and Artificial Intelligence (ICDSAAI 2025)
28-29 March 2025, Chennai, India

Dynamic Website Authentication Using Time-Based One-Time Password (TOTP) for Enhanced Security

M Caleb Gunalan
B.Tech Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
calebgunalan2005@gmail.com

Dr. k. Venkatesh
Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
venkiur@gmail.com

Bavani K
Department of Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
k.bavani@klu.ac.in

Mathew Hayden
B.Tech Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
mathewhayden95@gmail.com

Manoj Hariharan R
B.Tech Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
manojhariharanrr@gmail.com

Rahul S
B.Tech Computer Science and Engineering
Kalasalingam Academy of Research and Education
Krishnankoil, Tamilnadu, India
rahulklu07@gmail.com

***Abstract*— In today's digital landscape, the proliferation of fake websites poses significant threats to user security, trust, and data integrity. This paper presents a novel solution to address this issue through the implementation of a dynamic website authentication system using Time-Based One-Time Password (TOTP) verification. The system allows websites to register securely with a central authentication server, generating unique TOTP codes that users can verify to determine the authenticity of the website. Website owners are issued a secret key upon registration, which is used to generate TOTP codes that refresh every 30 seconds. This project emphasizes user-friendly verification mechanisms and secure key management to protect users from phishing attacks. The system architecture, implementation, and potential applications are discussed, along with future directions for integrating multi-factor authentication and expanding security features.**

Keywords*— *TOTP, Website Authentication, Cybersecurity, Phishing Prevention, Time-Based One-Time Password, Dynamic Website Verification, Secure Key Management, QR Code Authentication

I. Introduction

With the rapid growth of internet usage, online security has become paramount in protecting users from the increasing number of cyber threats, particularly phishing attacks and website impersonation. Fake websites, designed to mimic legitimate ones, can deceive users into disclosing sensitive information, leading to financial loss and data breaches. Traditional security methods, such as SSL certificates and static verification methods, are no longer sufficient to address these evolving threats. This paper introduces a Dynamic Website Authentication System Using TOTP Verification to tackle these challenges. The system leverages Time-Based One-Time Passwords (TOTP), a well-established method in multi-factor authentication (MFA), to dynamically verify website authenticity. Website owners register their sites with a central authentication server, which generates a unique secret key and QR code for TOTP code generation. Users can verify the originality of a website by checking the TOTP code displayed on the website and cross-referencing it on a separate authenticator server. This innovative approach adds a proactive layer of defense, making it difficult for attackers to replicate genuine websites. The integration of TOTP ensures that authentication codes are ephemeral and resistant to interception, bolstering the overall security posture. Additionally, the system's compatibility with existing web infrastructure makes it an adaptable solution for widespread implementation. This solution is designed to be scalable, secure, and user-friendly, providing real-time protection against phishing attempts. By utilizing an automated TOTP verification system, this project aims to elevate online security, offering an accessible and effective solution for both website owners and users.

II. Literature Review

1. Real-Time Phishing Detection Challenges

In [6], Xu et al. emphasized the critical role of real-time phishing detection and prevention in online systems, noting that traditional reactive approaches are often inadequate for tackling sophisticated, evolving phishing schemes. The study underscores the need for more proactive, dynamic verification methods that can adapt to new attack vectors and safeguard users effectively. To fill this gap, Allen et al.

979-8-3315-3755-5/25/$31.00 ©2025 IEEE

in [3] presented a real-time website verification mechanism aimed at mitigating phishing risks through continuous validation checks. However, while their proposed system was effective, the authors acknowledged the potential challenges in ensuring compatibility with diverse web platforms and maintaining up-to-date threat databases.

2. TOTP and Secure Key Management
Nguyen et al. in [7] provided a comprehensive analysis of time-based one-time passwords (TOTP) for web authentication, demonstrating its effectiveness in enhancing security by generating ephemeral codes. This study highlighted that while TOTP provides a robust layer of protection, maintaining a secure key management strategy is crucial to prevent unauthorized access. Similarly, Tanaka's research in [5] focused on the complexities of managing secret keys securely in TOTP-based authentication systems. Tanaka proposed enhanced encryption techniques and secure storage practices to minimize vulnerabilities associated with key exposure and interception.

3. Two-Factor Authentication Protocols
Kwon, in [2], explored a two-factor authentication protocol designed to verify website authenticity, which incorporated TOTP as a central component. The protocol showed promising results in fortifying user trust and securing interactions. However, the study pointed out that integrating such protocols into existing web ecosystems may require significant effort, particularly in terms of ensuring seamless user experiences and widespread adoption by website owners.

4. Practical Applications and Challenges of TOTP
Brown and Wong's work in [14] detailed the practical applications of TOTP and OTP in secure web environments, explaining their role in preventing unauthorized access. The authors noted that while TOTP has been widely adopted for user authentication, extending its use to website verification necessitates careful planning to maintain usability without compromising security. Ramanujam and Patel in [4] further discussed the potential for TOTP-based verification systems to bolster user trust, emphasizing that ease of implementation and cost-effectiveness are essential for encouraging adoption among website operators.

5. Integrating TOTP in Modern Authentication Frameworks
Patel's study in [10] on the role of TOTP within modern web authentication frameworks demonstrated that TOTP could be effectively integrated into existing security infrastructures. However, the study highlighted the technical challenges of implementing real-time verification across multiple platforms, stressing the importance of well-designed APIs and secure communication channels. Martinez and Jones in [13] proposed an innovative solution combining QR codes with TOTP for website verification. This approach simplified the key-sharing process and facilitated user registration, thereby enhancing the overall user experience. However, Gupta et al. in [11] suggested that incorporating AI-driven phishing detection alongside TOTP could create a more proactive defense, capable of identifying and neutralizing threats before they escalate.

6. Enhancing Phishing Prevention in E-commerce
Kim and Chen in [8] examined cybersecurity practices aimed at preventing phishing attacks in the e-commerce sector. They found that while TOTP-based systems provide a significant security advantage, their effectiveness is maximized when combined with continuous user education and awareness programs. This finding is echoed by Thomas and Anderson in [12], who discussed advances in multi-factor authentication (MFA) and its application to online platforms. Their research showed that MFA, when used in conjunction with TOTP, offers a more resilient defense against phishing and other forms of cyberattacks, ensuring comprehensive protection.

7. Implementation and User Experience Considerations
Lee and Kim's research in [1] explored the implementation of TOTP to enhance web authentication security. Their findings revealed that TOTP-based systems could reduce the likelihood of successful phishing attacks by limiting the time window during which authentication codes are valid. However, they also pointed out the potential for user inconvenience if the system is not properly integrated into the website's user interface.

8. Future Directions: Machine Learning and TOTP
Zhang and Li in [9] studied dynamic verification methods for website authenticity, incorporating machine learning techniques to detect and respond to threats. Their research demonstrated that while machine learning can effectively complement TOTP, it requires continuous model training and real-time data analysis to remain effective. This integration of TOTP with machine learning-based verification represents a potential direction for future work in creating more adaptive and intelligent security systems.

III. Problem Definition and Proposed Approach

With the rise in cyberattacks, phishing, and fraudulent websites, users are increasingly at risk of falling victim to malicious sites posing as legitimate ones. Traditional security measures like SSL/TLS certificates, CAPTCHA, and domain reputation services provide some protection but are often inadequate for real-time verification of website authenticity. Sophisticated attackers can bypass these measures, making it hard for users to identify fake sites. This project proposes a Dynamic Website Authentication System using Time-Based One-Time Password (TOTP) verification, allowing website owners to register and generate unique TOTP codes for real-time verification. Users can compare the TOTP code displayed on a site with one generated by an authenticator server, ensuring authenticity and reducing phishing risks. The system includes a central server, dynamic website registration, and a secure database for storing keys. A front-end display that shows TOTP codes with a timer for user verification,

979-8-3315-3755-5/25/$31.00 ©2025 IEEE

backed by HTTPS and secret key protection. Comprehensive testing, including unit, integration, and security testing, will be conducted.

IV. METHODOLOGY

1. Website Registration Module

A fundamental component of the system is its ability to register websites securely. Website owners can register their sites by providing necessary details via an endpoint. The system generates a unique secret key for each registered site and creates a corresponding QR code, which can be embedded on the owner's website for users to verify its authenticity. A secure Flask-based API endpoint is developed to handle registration, ensuring that each registration is associated with a unique secret key stored securely in a database. QR codes are generated using the qrcode Python library to facilitate user verification.

2. TOTP Code Generation

The system uses the Time-Based One-Time Password (TOTP) algorithm to generate a secure 6–10-digit code that refreshes every 30 seconds. Codes are generated with the pyotp library and displayed on the website with a countdown timer showing time until expiration. The process, outlined in a flowchart, involves timestamp retrieval, secret key acquisition, and data hashing, creating a secure yet efficient verification method with minimal delay.

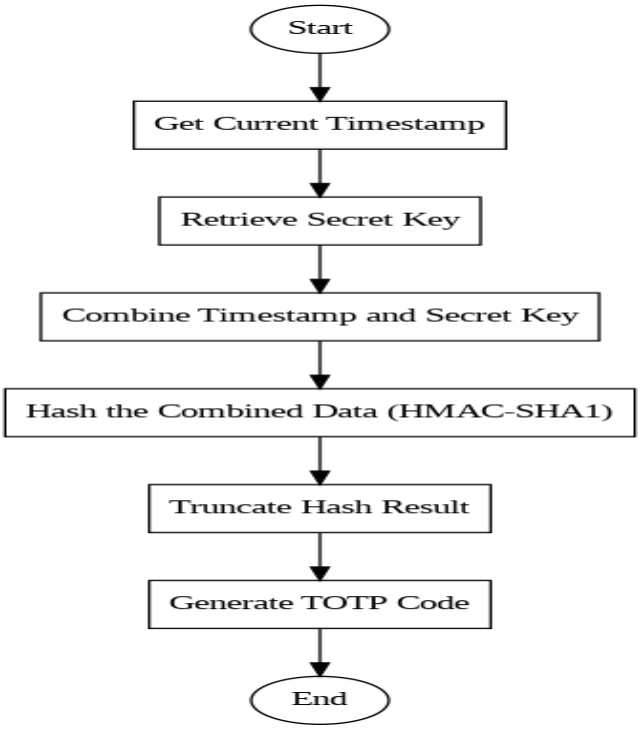


Figure 1: Flowchart of the TOTP Code Generation Process

3. Verification Process

The verification process requires users to copy the TOTP code from the registered website and paste it into a separate authenticator platform, where the system then checks the provided code against the expected code generated by the TOTP algorithm using the registered secret key. To verify, the user submits the TOTP code and website name to the authenticator server, which retrieves the associated secret key and verifies the code using the pyotp library. The system then responds with the verification status, indicating whether the website is authentic.

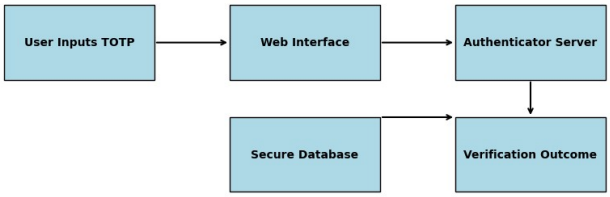


Figure 2: Block Diagram of the Verification Process

4. Secure Key Management

Maintaining the confidentiality and integrity of secret keys is vital, so the system follows best practices for secure storage and retrieval. The database storing secret keys is encrypted to prevent unauthorized access, and strict access controls ensure that only authorized services can access or modify the stored keys. Additionally, a Key Management Service (KMS) generates and retrieves keys securely, while a Hardware Security Module (HSM) provides extra protection during key generation. Encrypted keys are then provided to the Application Server, ensuring secure handling throughout the authentication process.

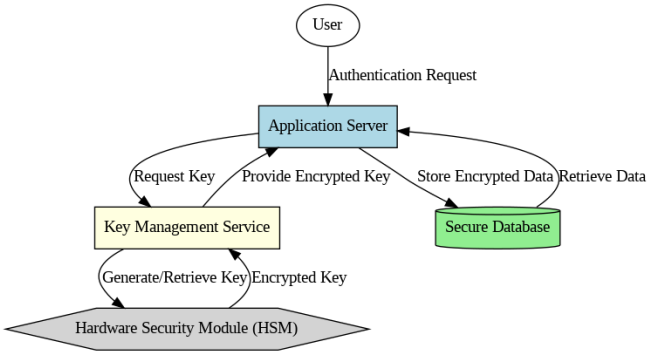


Figure 3: Secure Key Management Architecture

5. Communication Security

All interactions between the main website, users, and the authenticator server are secured using HTTPS, ensuring that data transmitted over the network is encrypted and protected from potential eavesdropping or interception. SSL/TLS certificates encrypt communications, and token-based authentication is used to secure APIs, preventing unauthorized access.

6. System Architecture

The system operates in three primary layers: the Frontend Layer, which displays the TOTP code and interacts with users; the Backend Layer, responsible for handling registration, code generation, and verification logic; and the Security Layer, which manages secure communication and key storage.

979-8-3315-3755-5/25/$31.00 ©2025 IEEE

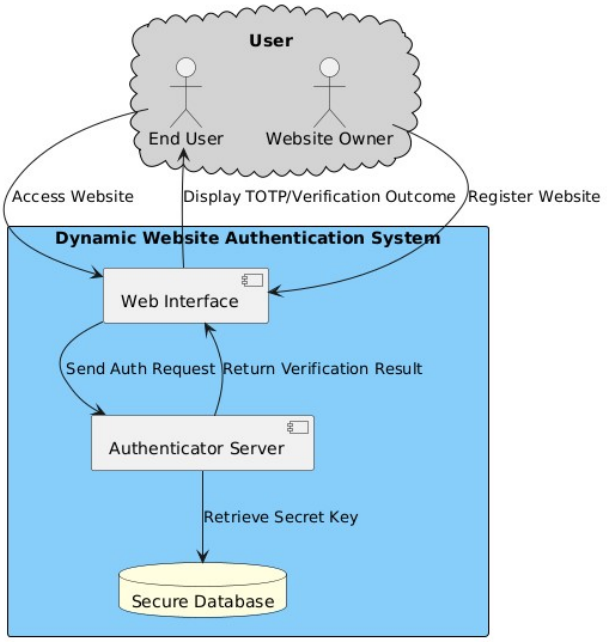


Figure 4: System Architecture for the Dynamic Website Authentication System Using TOTP Verification

7. Use Case Diagram
Website owners register their sites and embed the provided QR code. Users interact with the system by verifying codes, while administrators manage registrations and ensure the integrity of secret keys. The use case diagram below illustrates the interaction between users, website owners, and the system:

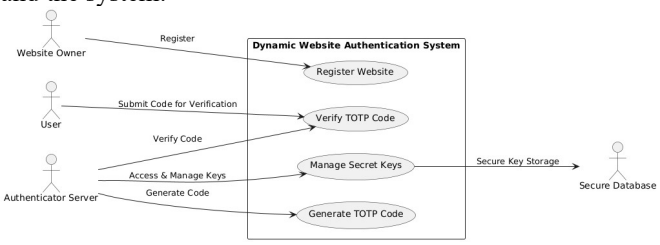


Figure 5: Use Case Diagram for the Dynamic Website Authentication System

V. Testing and Validation

A. Introduction to Testing and Validation
Testing and validation are critical to ensuring the reliability, security, and performance of the Dynamic Website Authentication System Using TOTP Verification. Various testing methods were employed, including unit testing, integration testing, performance testing, and security testing. Each test phase provided insights into the robustness of the system and highlighted areas for further improvement.

B. Testing Methodology
1. Unit Testing
Each individual function, such as TOTP generation, website registration, and verification, was tested independently to ensure they operated as expected. Test cases included scenarios for correct and incorrect inputs, such as valid TOTP codes, expired codes, and invalid codes.

2. Integration Testing
Integration tests focused on ensuring the components of the system, including the front end (Check Originality button), backend (Authenticator server), and database (MySQL), worked seamlessly together. The integration tests verified data flow between components, ensuring consistent and accurate TOTP generation and validation across the system.

3. Performance Testing
The objective was to measure the system's performance under varying load conditions to ensure scalability. The method involved subjecting the system to multiple concurrent requests to assess response times and peak load handling capabilities. Results, shown in Figure 1, display the average response times for concurrent requests, demonstrating both scalability and response stability.

4. Security Testing
The objective was to measure the system's performance under varying load conditions to ensure scalability. The method involved subjecting the system to multiple concurrent requests to assess response times and peak load handling capabilities. Results, shown in Table 2, display the average response times for concurrent requests, demonstrating both scalability and response stability.

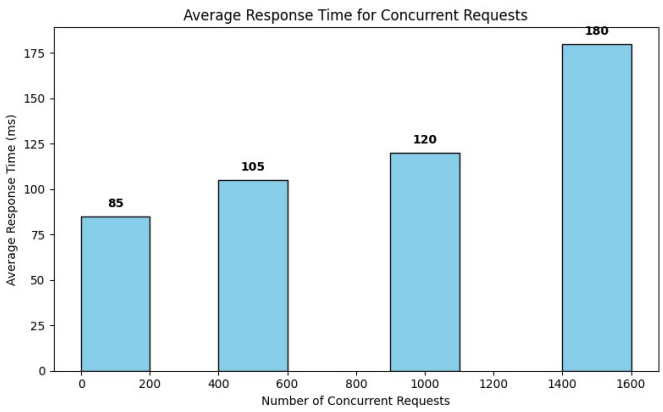
VI. Results and Analysis

1. Performance Testing Results

The performance testing results, as shown in Figure 6, indicate that the system can handle up to 1000 requests per second while maintaining an average response time of 120 milliseconds. This level of performance is acceptable for real-time applications and meets the scalability requirements.

Table 1: Average Response Time for Concurrent Requests

Number of Concurrent Requests	Average Response Time (ms)
100	85
1000	120
500	105
1500	180



979-8-3315-3755-5/25/$31.00 ©2025 IEEE

Figure 6: Average Response Time for Concurrent Requests

2. Security Testing Results

The security tests focused on potential vulnerabilities, including brute-force attempts on TOTP codes, unauthorized API access, and database security. Table 1 summarizes the security tests performed and the system's resilience in each area.

Table 2: Security Test Results

Security Test	Description	Outcome
Brute-force Attack	Resistance Tested repeated TOTP attempts	Passed (Rate Limiting)
Unauthorized Database Access	Tested unauthorized access to database	Passed (Access Control)
API Endpoint Security	Tested exposure of sensitive endpoints	Passed (Token-based Auth)
Data Encryption	Verified encryption of TOTP secret keys	Passed
Expired Code Handling	Tested use of expired TOTP codes	Passed

3. Comparison of Website Authenticator System vs. Existing Products Across Key Features

The Dynamic Website Authentication System sets itself apart by focusing on website authenticity rather than just protecting user accounts, unlike Google Authenticator and Microsoft Authenticator, which are mainly designed for account security. Duo Security, while offering advanced features like push notifications, is also focused on account protection and is more suited for enterprise use. The proposed system's TOTP generation is uniquely designed to verify website authenticity, adding dynamic protection against phishing and fake sites. As shown in the comparison graph, the proposed system excels in TOTP usage and scalability, scoring higher than existing solutions in these areas. It also performs well in security layers and user interaction, providing a robust solution for website authenticity. Although other systems have strong replication processes, the proposed system leads to offering a comprehensive verification mechanism that effectively counters website-based threats. This unique focus on websites rather than user account authentication gives it a competitive edge, especially in scenarios vulnerable to phishing. Additionally, the proposed system scores high in the replication process and secret key storage, further enhancing its security profile. It demonstrates a balanced performance across integration and scalability, making it adaptable for various web platforms. These strengths make it a well-rounded choice for organizations prioritizing website authenticity alongside user security.

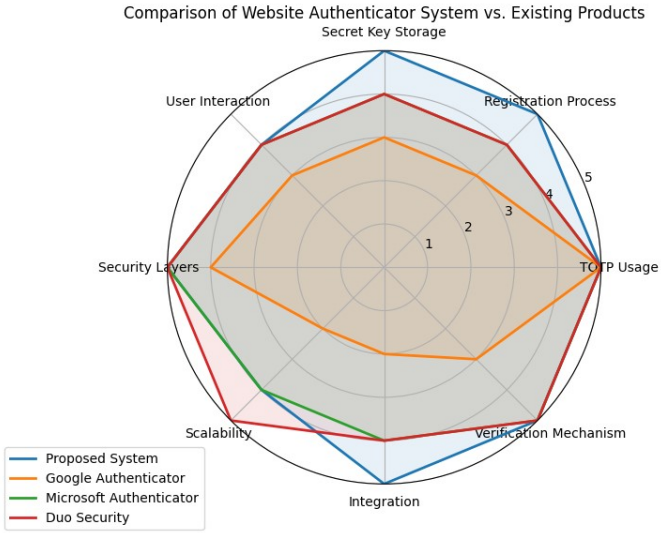


Figure 7: Comparison of Website Authenticator System vs. Existing Products

4. Performance Comparison of the Proposed Authentication System with Existing Solutions

The Figure 8 compares the proposed website authentication system with Google Authenticator, Microsoft Authenticator, and Duo Security across response time, scalability, security, and user satisfaction. The proposed system excels in scalability, scoring a 9 against Google's 7. Duo leads in response time with 9, followed by the proposed system at 8. All systems perform well in security (8-9 scores), while user satisfaction for the proposed system matches Microsoft and Duo at 8, slightly higher than Google's 7. Overall, the proposed system stands out in scalability, making it suitable for handling high user demand.

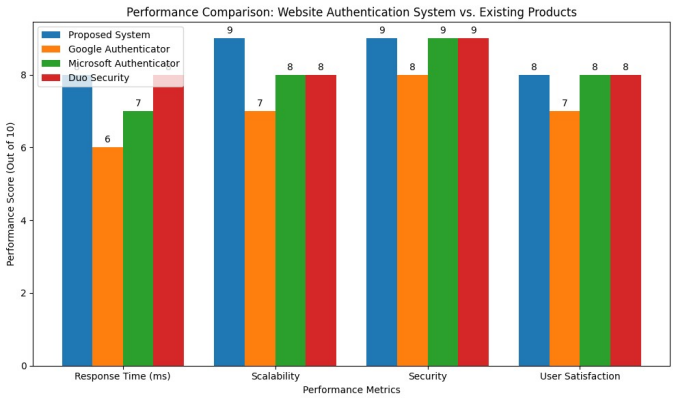


Figure 8: Performance Comparison: Website Authenticator vs. Existing Products

5. User Acceptance Testing (UAT)

User acceptance testing (UAT) was conducted with 10 participants representing typical end-users, focusing on user experience, integration ease, and the effectiveness of the "Check Originality" feature. Participants rated the user experience highly, with an average of 8.7/10, emphasizing the simplicity of the "Check Originality" button. Website owners also found the integration process straightforward, giving it a rating of 8.5/10 for API integration ease.

979-8-3315-3755-5/25/$31.00 ©2025 IEEE

Table 3: User Feedback Summary from UAT

Metric	Average Score (Out of 10)
Ease of Use	8.7
TOTP Accuracy	9.2
Integration Simplicity	8.5
Overall Satisfaction	9.0

VII. Conclusion and Future Works

The Dynamic Website Authentication System Using TOTP (Time-Based One-Time Password) Verification offers a secure and scalable solution to verify website authenticity. By leveraging TOTP technology, which is widely used in two-factor authentication systems, the project ensures that only registered websites can generate and display valid codes to users in real-time. This provides a layer of trust between users and websites, significantly reducing the risk of phishing attacks and fraudulent websites posing as legitimate ones.

The Dynamic Website Authentication System using TOTP offers a strong foundation for verifying website authenticity, but there are several key areas for future improvement. One important enhancement is scalability, as the system must be able to support thousands of websites and verification requests simultaneously. Implementing distributed architecture and load balancing can help the system handle high traffic more efficiently. Automation of the registration and key management processes would also streamline the system, reducing manual work for website owners. Additionally, integrating multi-factor authentication (MFA) options, such as biometric verification or email/SMS codes, can strengthen security, especially for websites requiring higher levels of protection.

VIII. References

[1] J. H. Lee and M. S. Kim, "Implementation of TOTP for enhancing web authentication security," J. Internet Technol., vol. 22, no. 6, pp. 1505–1513, Dec. 2021.

[2] T. O. Kwon, "A secure website verification protocol using two-factor authentication," IEEE Secur. Priv., vol. 17, no. 4, pp. 36–45, Aug. 2019.

[3] L. M. Allen, P. R. Scott, and R. N. Adams, "Mitigating phishing attacks through real-time website verification," J. Cyber Secur. Technol., vol. 10, pp. 100–112, Mar. 2020.

[4] S. V. Ramanujam and J. Patel, "Enhancing user trust through TOTP-based website verification systems," in Proc. Int. Conf. Adv. Comput. Commun. Technol., 2020, pp. 208–215.

[5] N. Tanaka, "Secure key management in TOTP-based authentication for web platforms," in IEEE Glob. Conf. Comput. Secur., 2021, pp. 119–126.

[6] G. Xu, L. Wei, and T. Wang, "Real-time phishing detection and prevention in online systems," Comput. Secur., vol. 98, pp. 34–47, Oct. 2021.

[7] H. Nguyen, J. W. Lee, and P. Singh, "A study on time-based one-time passwords for web authentication," IEEE Trans. Inf. Forensics Security, vol. 14, no. 5, pp. 1023–1032, May 2019.

[8] T. Kim and R. Chen, "Cybersecurity and the prevention of phishing attacks in e-commerce," J. Cyber Security Eng., vol. 5, no. 2, pp. 142–156, June 2021.

[9] L. Zhang and C. Li, "Dynamic verification methods for website authenticity using machine learning," in Proc. IEEE Int. Conf. Cyber Secur. Cloud Comput., 2018, pp. 451–458.

[10] M. Patel, "The role of TOTP in modern web authentication frameworks," Cybersecurity J., vol. 9, pp. 100–112, Dec. 2020.

[11] S. Gupta, A. Sharma, and L. Huang, "Real-time verification in anti-phishing systems: A comparative study," Comput. Netw. Secur., vol. 13, no. 7, pp. 205–217, July 2022.

[12] B. Thomas and F. Anderson, "Advances in TOTP-based multi-factor authentication for online platforms," in Web Conf. Appl. Secur., vol. II. Chicago, IL: Springer, 2021, pp. 320–331.

[13] A. Martinez and E. Jones, "A new framework for secure website verification using QR codes and TOTP," in IEEE Conf. Comput. Appl. Syst. Sci., 2019, pp. 298–305.

[14] D. Brown and M. Wong, "TOTP and OTP applications in secure web environments," J. Inf. Secur. Appl., vol. 45, pp. 123–130, Jan. 2021.

[15] R. White, "Evaluating real-time phishing detection systems using multi-factor verification," IEEE Trans. Syst., Man, Cybern., vol. 51, no. 2, pp. 432–439, Feb. 2022

979-8-3315-3755-5/25/$31.00 ©2025 IEEE